

AI Engineer Technical Assessment

Estimated Time: 2–3 hours

Assessment Type: Unpaid (used only to evaluate technical ability)

Objective

Develop a prototype platform (with or without a graphical user interface) that accepts a floor plan and generates photorealistic interior design renders using products from a supplied furniture and renovation catalog. The platform should demonstrate strong AI engineering, system design, retrieval, and image generation capabilities.

Functional Requirements

- Accept floor plans in PNG, JPG, JPEG, or PDF format.
- Analyze the layout and detect rooms.
- Allow users to select an interior design style before generation.
- Generate multiple photorealistic renders from different viewpoints.
- Use only products from the supplied furniture and renovation catalog.
- Return the products used in every render.
- Provide a REST API, CLI, or minimal web interface.

User Features

- Choose an interior design style (Scandinavian/Nordic, Japanese, Japandi, Modern, Contemporary, Minimalist, Industrial, Classic, Luxury, Rustic, Bohemian, etc.).
- Choose a preferred color palette.
- Generate multiple design variations.
- Regenerate designs while preserving the same scene unless the user requests changes.

Scene Consistency Requirements

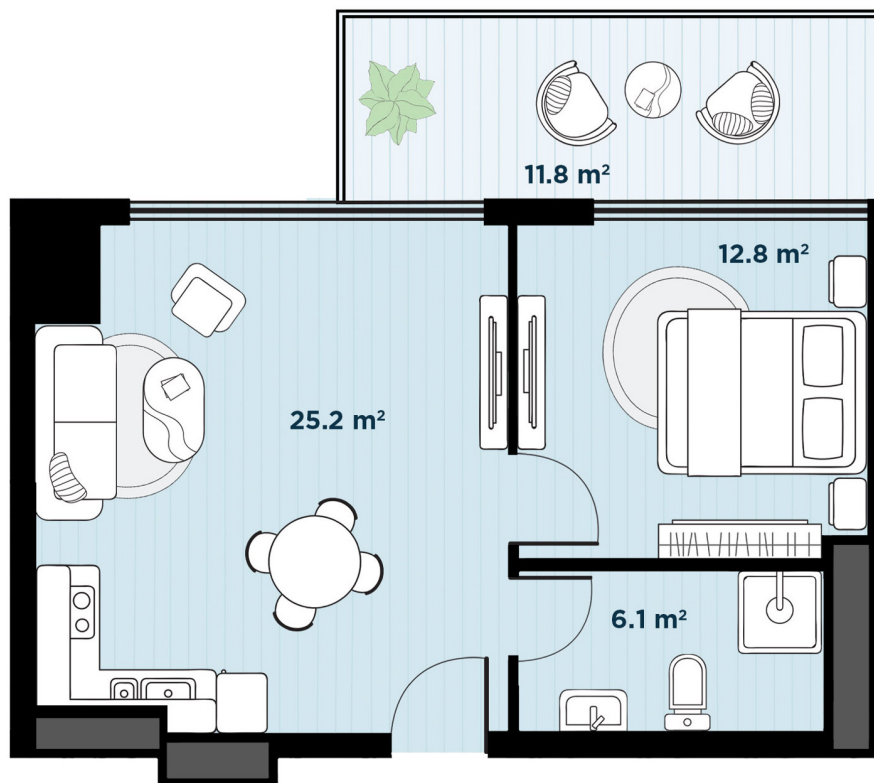
- Every render must represent the exact same interior scene from different camera positions.
- The room layout must remain identical across every render.
- Furniture, decorations, lighting fixtures, rugs, artwork, plants, curtains, and accessories must remain in the same locations.
- Objects must not move, disappear, appear, rotate, resize, or change identity between viewpoints.
- Walls, floors, ceilings, doors, windows, and architectural elements must remain unchanged.
- Only the camera viewpoint may change.
- Furniture proportions and room dimensions should closely match the floor plan.
- Avoid mismatched layouts, floating objects, inconsistent geometry, or duplicated/missing items.

Sample Product Catalog

For demonstration purposes, candidates may use products from the following sample catalogs:

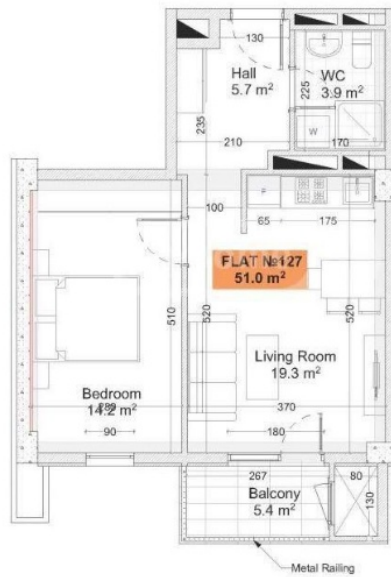
- Gorgia – furniture, renovation materials, lighting, flooring, paint, bathroom and kitchen products.
- Comforter – sofas, tables, beds, wardrobes, office furniture, mattresses, textiles and home accessories.
- Design the solution so catalogs can be imported and indexed.
- Support searching products by category, dimensions, color, material, style, and price.
- Every visible furniture or renovation element should correspond to a real catalog product whenever possible.
- Return product names/IDs (or URLs), quantities, and estimated total cost.
- The architecture should support multiple suppliers and allow replacing catalogs with minimal changes.

Sample Floor Plans





კორპუსი №2
სართული 19
ბინა №127
 $45.6+5.4=51.0 \text{ m}^2$
BS1-45-E-T



Evaluation Criteria

- Overall AI engineering and architecture.
- Model selection and workflow design.
- Code quality and documentation.
- Photorealism.
- Style adherence.
- Scene consistency across viewpoints.
- Layout preservation.
- Furniture placement consistency.
- Object identity preservation.
- Proportion and geometry accuracy.
- Correct usage of supplied product catalogs.
- Scalability and production readiness.